

Li-Ion Cell Charge Manegement



File: tracker_board_cell_charge_magagement.kicad_sch

Power Rails



File: tracker_board_power_rails.kicad_sch

Host_MCU



File: host_mcu.kicad_sch

NB_IoT_Modem_p1

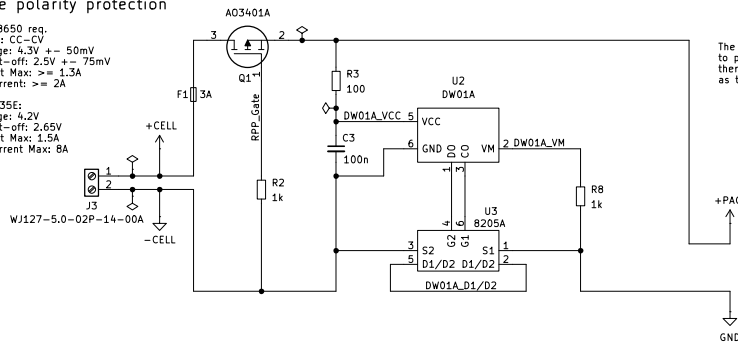


File: lte_modem.kicad_sch

Single Cell Li-Ion/Li-Pol Charge Protection with reverse polarity protection

Li-Ion-Akku 18650 req.
Charge-Method: CC-CV
Charging Voltage: 4.3V $\pm$ 50mV
Discharging Cut-off: 2.5V $\pm$ 75mV
Charging Current Max: $\geq$ 1.3A
Discharging Current: $\geq$ 2A

e.g. INR18650-35E:
Charging Voltage: 4.2V
Discharging Cut-off: 2.65V
Charging Current Max: 1.5A
Discharging Current Max: 8A

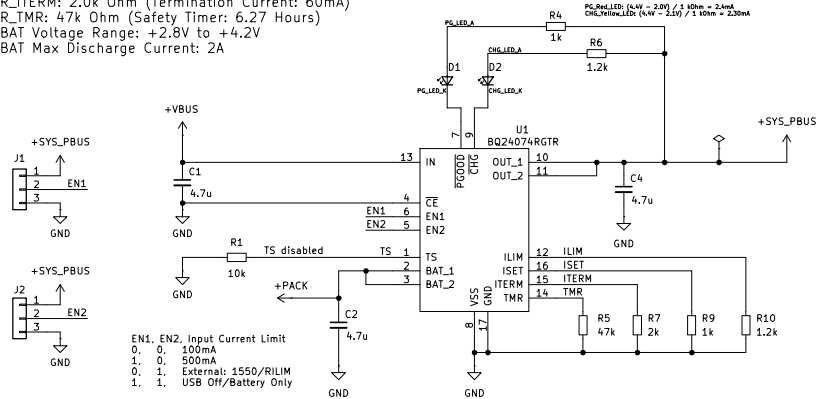


The DW01A is intended as a last resort in rare cases to prevent the cell from being damaged or catching fire; therefore, small tolerances in the cell values are acceptable as this only happens rarely.

Pack Ratings:
Voltage Range: +2.5V to +4.2V
Max Charge current: 1.3A
Max Discharge current: 2A

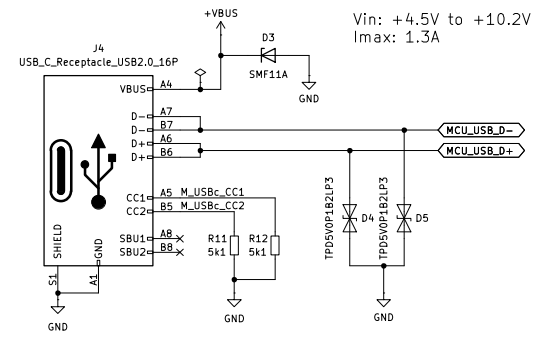
Single-Cell Li-Ion/Li-Pol Charger & Power Management:

Input Voltage Range: +4.5V to +10.2V
The System Output Voltage: +4.4V with input power.
Without input power, it tracks the battery voltage (2.8V to 4.2V).
R_ILIM: 1.2k Ohm (Input Limit: 1.29A)
R_ISET: 1.0k Ohm (Charge Current: 890mA)
R_TERM: 2.0k Ohm (Termination Current: 60mA)
R_TMR: 47k Ohm (Safety Timer: 6.27 Hours)
BAT Voltage Range: +2.8V to +4.2V
BAT Max Discharge Current: 2A

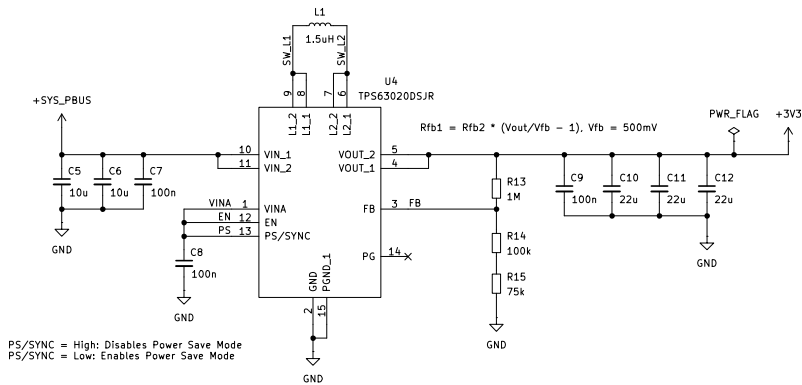


Main Power-Supply & Data Connector:

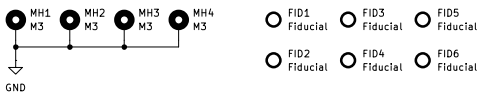
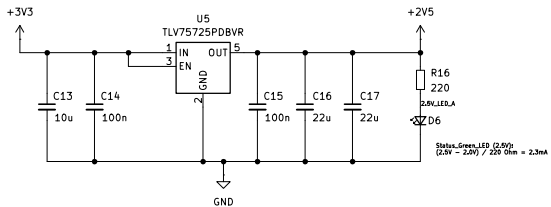
Vin: +4.5V to +10.2V
Imax: 1.3A

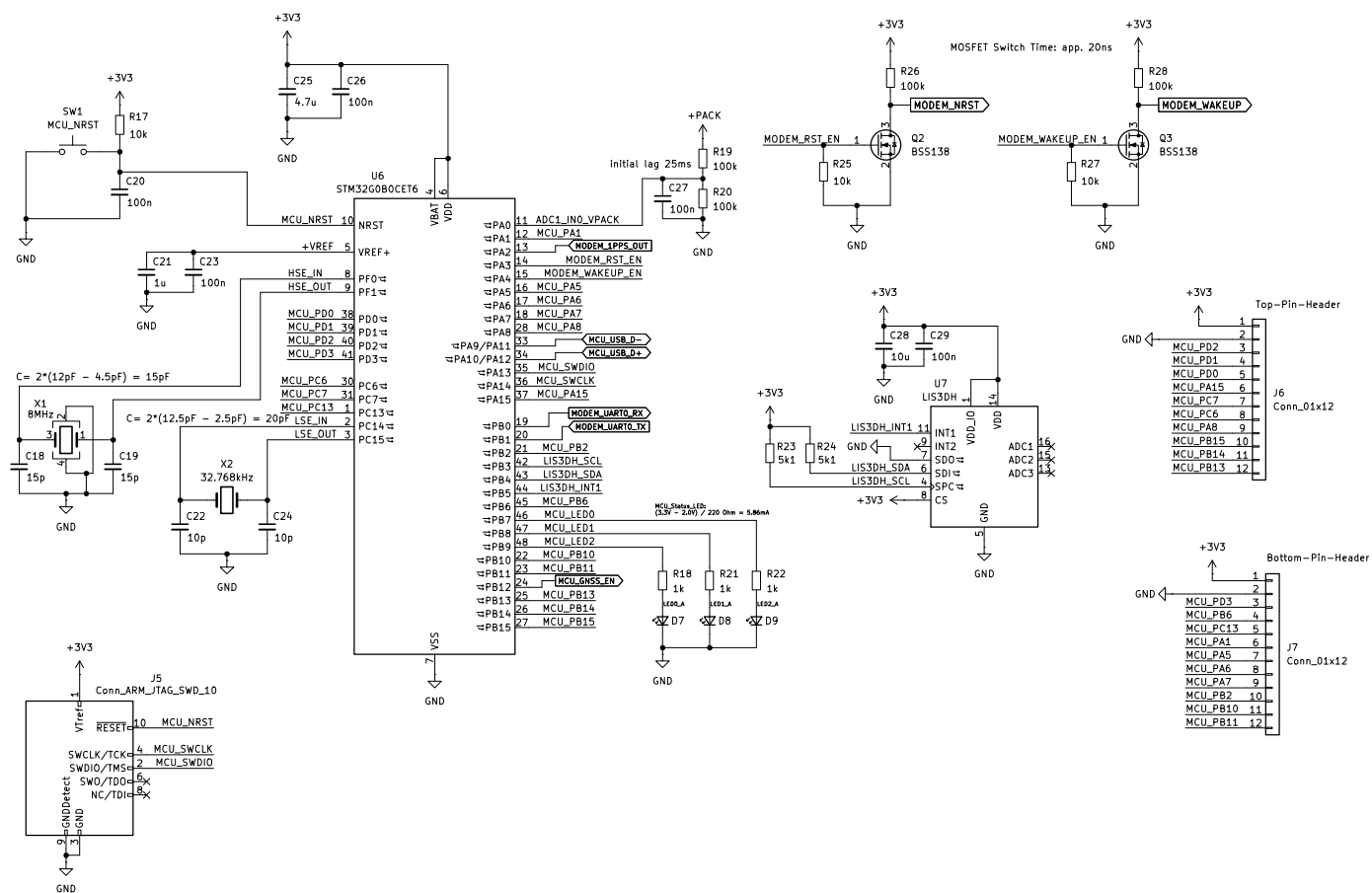


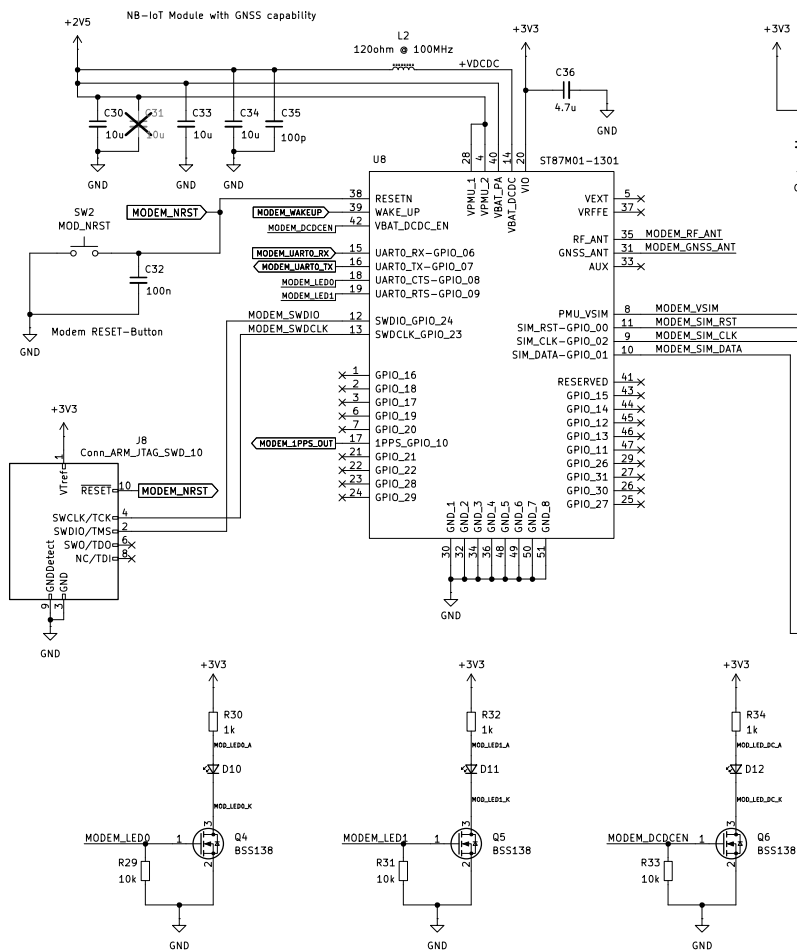
Buck-Boost Converter:
 Input voltage range: 2.8V to 4.4V
 Output voltage: +3.3V
 $I_{out_max} = 2A$ in Boost-Mode, 4A in Buck-Mode



LDO Regulator:
 Input voltage: +3.3V
 Output voltage: +2.5V
 $I_{out_max} = 1A$







GNSS_ANT Power (3.3V Supply):
 Startup: $(0.38 \times 1nF + 0.07) = 0.45 \text{ V/ms} \rightarrow 3.3V / 0.45 = 7.33ms$
 Shutdown: $2.2 \times (1037 \text{ Ohm} \times 100nF) = 0.228ms$
 Peak Discharge Current: $3.3V / 1037 \text{ Ohm} = 3.18mA$

